

Министерство науки и высшего образования Российской Федерации
НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ
ТОМСКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ (НИ ТГУ)
Институт прикладной математики и информатики

ОТЧЕТ ПО ЛАБОРАТОРНОЙ РАБОТЕ
по дисциплине «Системы управления базами данных»

JOINS

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подпись

«26» ноября 2022 г.

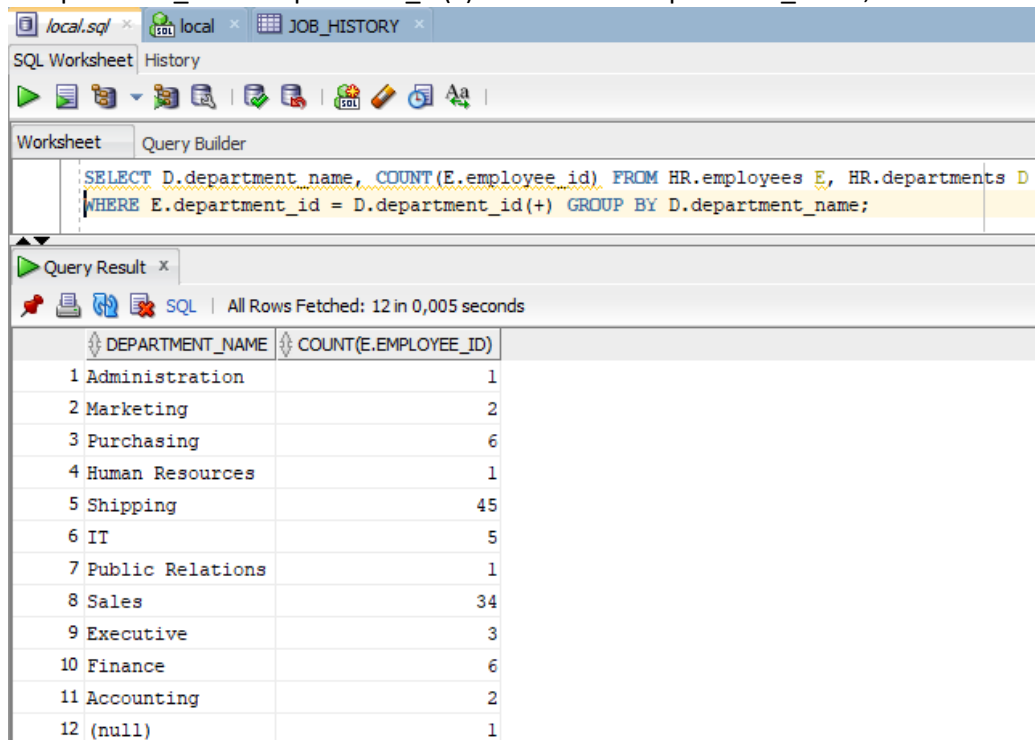
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«26» ноября 2022 г.

1. Вывести название отделения и количество сотрудников в каждом отделении.

SELECT D.department_name, COUNT(E.employee_id) FROM HR.employees E, HR.departments D WHERE E.department_id = D.department_id(+) GROUP BY D.department_name;



The screenshot shows an SQL Worksheet with the following query:

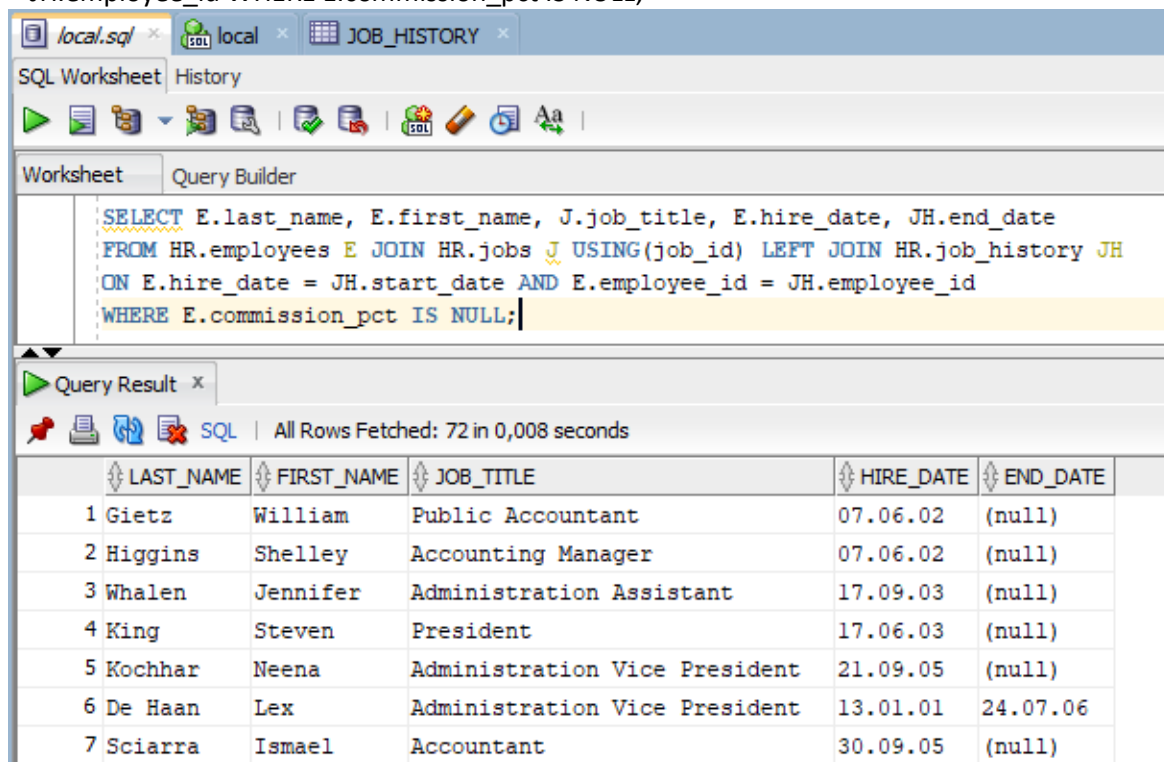
```
SELECT D.department_name, COUNT(E.employee_id) FROM HR.employees E, HR.departments D
WHERE E.department_id = D.department_id(+) GROUP BY D.department_name;
```

The Query Result shows 12 rows of data:

DEPARTMENT_NAME	COUNT(E.EMPLOYEE_ID)
1 Administration	1
2 Marketing	2
3 Purchasing	6
4 Human Resources	1
5 Shipping	45
6 IT	5
7 Public Relations	1
8 Sales	34
9 Executive	3
10 Finance	6
11 Accounting	2
12 (null)	1

2. Вывести фамилию, имя, должность, промежуток работы для сотрудников, которые работают без комиссионных.

SELECT E.last_name, E.first_name, J.job_title, E.hire_date, JH.end_date FROM HR.employees E JOIN HR.jobs J USING(job_id) LEFT JOIN HR.job_history JH ON E.hire_date = JH.start_date AND E.employee_id = JH.employee_id WHERE E.commission_pct IS NULL;



The screenshot shows an SQL Worksheet with the following query:

```
SELECT E.last_name, E.first_name, J.job_title, E.hire_date, JH.end_date
FROM HR.employees E JOIN HR.jobs J USING(job_id) LEFT JOIN HR.job_history JH
ON E.hire_date = JH.start_date AND E.employee_id = JH.employee_id
WHERE E.commission_pct IS NULL;
```

The Query Result shows 7 rows of data:

LAST_NAME	FIRST_NAME	JOB_TITLE	HIRE_DATE	END_DATE
1 Gietz	William	Public Accountant	07.06.02	(null)
2 Higgins	Shelley	Accounting Manager	07.06.02	(null)
3 Whalen	Jennifer	Administration Assistant	17.09.03	(null)
4 King	Steven	President	17.06.03	(null)
5 Kochhar	Neena	Administration Vice President	21.09.05	(null)
6 De Haan	Lex	Administration Vice President	13.01.01	24.07.06
7 Sciarra	Ismael	Accountant	30.09.05	(null)

3. Вывести фамилию, имя, зарплату сотрудников, работающих в отделениях, находящихся в Лондоне.

SELECT E.last_name, E.first_name, E.salary FROM HR.employees E JOIN HR.departments D
USING(department_id) JOIN HR.locations L USING(location_id) WHERE L.city = 'London';

WorksheetQuery Builder

```
SELECT E.last_name, E.first_name, E.salary FROM HR.employees E JOIN HR.departments D
USING(department_id) JOIN HR.locations L USING(location_id) WHERE L.city = 'London';
```

Query Result x

    SQL

All Rows Fetched: 1 in 0,006 seconds

	LAST_NAME	FIRST_NAME	SALARY
1	Mavris	Susan	6500

4. Вывести id сотрудника, должность и количество проработанных дней для сотрудников из отделения 80.

SELECT E.employee_id, J.job_title, TRUNC(SYSDATE - E.hire_date) FROM HR.employees E JOIN HR.jobs J
USING(job_id) WHERE E.department_id = 80;

Worksheet

Query Builder

SELECT E.employee_id, J.job_title, TRUNC(SYSDATE - E.hire_date)

FROM HR.employees E JOIN HR.jobs J USING(job_id) WHERE E.department_id = 80;

▶

Query Result x

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SQL | All Rows Fetched: 34 in 0,005 seconds

	EMPLOYEE_ID	JOB_TITLE	TRUNC(SYSDATE-E.HIRE_DATE)
1	147	Sales Manager	6499
2	146	Sales Manager	6563
3	145	Sales Manager	6659
4	149	Sales Manager	5444
5	148	Sales Manager	5550
6	155	Sales Representative	5511

5. Вывести название отдела и фамилию, имя, город менеджера этого отдела.

SELECT d.department_name, E.last_name, E.first_name, L.city FROM HR.departments D JOIN
HR.employees E ON D.manager_id = E.employee_id JOIN HR.locations L USING(location_id);

Worksheet

Query Builder

```
SELECT d.department_name, E.last_name, E.first_name, L.city FROM HR.departments D
JOIN HR.employees E ON D.manager_id = E.employee_id JOIN HR.locations L USING(location_id);
```

Query Result x

| All Rows Fetched: 11 in 0,006 seconds

	DEPARTMENT_NAME	LAST_NAME	FIRST_NAME	CITY
1	Executive	King	Steven	Seattle
2	IT	Hunold	Alexander	Southlake
3	Finance	Greenberg	Nancy	Seattle
4	Purchasing	Raphaely	Den	Seattle
5	Shipping	Fripp	Adam	South San Francisco
6	Sales	Russell	John	Oxford
7	Administration	Whalen	Jennifer	Seattle
8	Marketing	Hartstein	Michael	Toronto
9	Human Resources	Mavris	Susan	London
10	Public Relations	Baer	Hermann	Munich
11	Accounting	Higgins	Shelley	Seattle

6. Вывести страну, город и название отдела, где работают хотя бы два сотрудника.

SELECT c.country_name, L.city, D.department_id FROM HR.departments D JOIN HR.locations L USING(location_id) JOIN HR.countries C USING(country_id) JOIN HR.employees E ON D.department_id = E.department_id GROUP BY c.country_name, L.city, D.department_id HAVING COUNT(E.employee_id) > 1;

Worksheet

Query Builder

```
SELECT c.country_name, L.city, D.department_id FROM HR.departments D
JOIN HR.locations L USING(location_id) JOIN HR.countries C USING(country_id)
JOIN HR.employees E ON D.department_id = E.department_id
GROUP BY c.country_name, L.city, D.department_id HAVING COUNT(E.employee_id) > 1;
```

Query Result x

All Rows Fetched: 8 in 0,003 seconds

	COUNTRY_NAME	CITY	DEPARTMENT_ID
1	United States of America	Southlake	60
2	United States of America	South San Francisco	50
3	United States of America	Seattle	90
4	United States of America	Seattle	100
5	United States of America	Seattle	110
6	United States of America	Seattle	30
7	Canada	Toronto	20
8	United Kingdom	Oxford	80

7. Вывести фамилию, имя, номер отдела и название отдела для каждого сотрудника.

SELECT E.last_name, E.first_name, E.department_id, D.department_name FROM HR.employees E LEFT JOIN HR.departments D ON E.department_id = D.department_id;

Worksheet

Query Builder

```
SELECT E.last_name, E.first_name, E.department_id, D.department_name
FROM HR.employees E LEFT JOIN HR.departments D ON E.department_id = D.department_id;
```

Query Result x

    SQL | All Rows Fetched: 107 in 0,005 seconds

LAST_NAME	FIRST_NAME	DEPARTMENT_ID	DEPARTMENT_NAME
1 Whalen	Jennifer	10	Administration
2 Hartstein	Michael	20	Marketing
3 Fay	Pat	20	Marketing
4 Raphaely	Den	30	Purchasing
5 Khoo	Alexander	30	Purchasing
6 Baida	Shelli	30	Purchasing
7 Tobias	Sigal	30	Purchasing

8. Вывести фамилию, имя, номер отдела и название отдела для каждого сотрудника из отдела 50 или 90.

SELECT E.last_name, E.first_name, E.department_id, D.department_name FROM HR.employees E LEFT JOIN HR.departments D ON E.department_id = D.department_id WHERE E.department_id IN (50,90);

Worksheet

Query Builder

```
SELECT E.last_name, E.first_name, E.department_id, D.department_name
FROM HR.employees E LEFT JOIN HR.departments D ON E.department_id = D.department_id
WHERE E.department_id IN (50,90);
```

Query Result x

All Rows Fetched: 48 in 0,003 seconds

	LAST_NAME	FIRST_NAME	DEPARTMENT_ID	DEPARTMENT_NAME
1	OConnell	Donald	50	Shipping
2	Grant	Douglas	50	Shipping
3	Weiss	Matthew	50	Shipping
4	Fripp	Adam	50	Shipping
5	Kaufling	Payam	50	Shipping
6	Vollman	Shanta	50	Shipping

9. Вывести название отдела, город и штат для каждого отдела.

SELECT D.department_name, L.city, L.state_province FROM HR.departments D JOIN HR.locations L
USING(location_id);

Worksheet		Query Builder
<pre>SELECT D.department_name, L.city, L.state_province FROM HR.departments D JOIN HR.locations L USING(location_id);</pre>		
Query Result x		
All Rows Fetched: 27 in 0,004 seconds		
DEPARTMENT_NAME	CITY	STATE_PROVINCE
1 IT	Southlake	Texas
2 Shipping	South San Francisco	California
3 Purchasing	Seattle	Washington
4 Administration	Seattle	Washington
5 Control And Credit	Seattle	Washington
6 Recruiting	Seattle	Washington

10. Вывести фамилию, имя, название отдела, город и штат для каждого сотрудника.

SELECT E.last_name, E.first_name, D.department_name, L.city, L.state_province FROM HR.employees E
LEFT JOIN HR.departments D ON E.department_id = D.department_id LEFT JOIN HR.locations L
USING(location_id);

Worksheet

Query Builder

```
SELECT E.last_name, E.first_name, D.department_name, L.city, L.state_province
FROM HR.employees E LEFT JOIN HR.departments D ON E.department_id = D.department_id
LEFT JOIN HR.locations L USING(location_id);
```

Query Result x

All Rows Fetched: 107 in 0,004 seconds

	LAST_NAME	FIRST_NAME	DEPARTMENT_NAME	CITY	STATE_PROVINCE
1	Hunold	Alexander	IT	Southlake	Texas
2	Ernst	Bruce	IT	Southlake	Texas
3	Austin	David	IT	Southlake	Texas
4	Pataballa	Valli	IT	Southlake	Texas
5	Lorentz	Diana	IT	Southlake	Texas
6	OConnell	Donald	Shipping	South San Francisco	California

11. Вывести фамилию, имя, название отдела, город и штат для каждого сотрудника, фамилия
которого содержит букву 'a'.

SELECT E.last_name, E.first_name, D.department_name, L.city, L.state_province FROM HR.employees E
LEFT JOIN HR.departments D ON E.department_id = D.department_id LEFT JOIN HR.locations L
USING(location_id) WHERE E.last_name LIKE '%a%';

SQL Worksheet | History



Worksheet | Query Builder

```
SELECT E.last_name, E.first_name, D.department_name, L.city, L.state_province
FROM HR.employees E LEFT JOIN HR.departments D ON E.department_id = D.department_id
LEFT JOIN HR.locations L USING(location_id) WHERE E.last_name LIKE '%a%';
```

Query Result x

 All Rows Fetched: 52 in 0,002 seconds

	LAST_NAME	FIRST_NAME	DEPARTMENT_NAME	CITY	STATE_PROVINCE
1	Pataballa	Valli	IT	Southlake	Texas
2	Grant	Douglas	Shipping	South San Francisco	California
3	Kaufling	Payam	Shipping	South San Francisco	California
4	Vollman	Shanta	Shipping	South San Francisco	California
5	Nayer	Julia	Shipping	South San Francisco	California
6	Landry	James	Shipping	South San Francisco	California

12. Получить все страны и соответствующие города, включая страны без каких-либо городов.

SELECT c.country_name, L.city FROM HR.locations L RIGHT OUTER JOIN HR.countries C ON L.country_id = C.country_id;

Worksheet	Query Builder
<pre>SELECT c.country_name, L.city FROM HR.locations L RIGHT OUTER JOIN HR.countries C ON L.country_id = C.country_id</pre>	
Query Result x All Rows Fetched: 34 in 0,004 seconds	
COUNTRY_NAME	CITY
1 Australia	Sydney
2 Brazil	Sao Paulo
3 Canada	Toronto
4 Canada	Whitehorse
5 Switzerland	Geneva
6 Switzerland	Bern

13. Получить все страны Африки и соответствующие города, в том числе страны без каких-либо городов.

SELECT C.country_name, L.city FROM HR.locations L RIGHT OUTER JOIN HR.countries C ON L.country_id = C.country_id RIGHT OUTER JOIN HR.regions R USING(region_id) WHERE R.region_name LIKE '%Africa%';

Worksheet	Query Builder
<pre>SELECT C.country_name, L.city FROM HR.locations L RIGHT OUTER JOIN HR.countries C ON L.country_id = C.country_id RIGHT OUTER JOIN HR.regions R USING(region_id) WHERE R.region_name LIKE '%Africa%'</pre>	
Query Result x All Rows Fetched: 6 in 0,006 seconds	
COUNTRY_NAME	CITY
1 Egypt	(null)
2 Nigeria	(null)
3 Kuwait	(null)
4 Zambia	(null)
5 Zimbabwe	(null)
6 Israel	(null)

14. Используйте синтаксис RIGHT JOIN для получения точно таких же результатов как в примере выше.

SELECT C.country_name, L.city FROM HR.locations L RIGHT JOIN HR.countries C ON L.country_id = C.country_id RIGHT JOIN HR.regions R USING(region_id) WHERE R.region_name LIKE '%Africa%';

Worksheet	Query Builder
<pre>SELECT C.country_name, L.city FROM HR.locations L RIGHT JOIN HR.countries C ON L.country_id = C.country_id RIGHT JOIN HR.regions R USING(region_id) WHERE R.region_name LIKE '%Africa%';</pre>	
Query Result x All Rows Fetched: 6 in 0,006 seconds	
COUNTRY_NAME	CITY
1 Egypt	(null)
2 Nigeria	(null)
3 Kuwait	(null)
4 Zambia	(null)
5 Zimbabwe	(null)
6 Israel	(null)

15. Несколько операций соединения в предложении FROM. Укажите номер и фамилию каждого сотрудника с номером сотрудника и фамилией его руководителя.

SELECT E1.employee_id, E1.last_name, E2.employee_id, E2.last_name FROM HR.employees E1 LEFT JOIN HR.employees E2 ON E1.manager_id = E2.employee_id;

Worksheet

Query Builder

```
SELECT E1.employee_id, E1.last_name, E2.employee_id, E2.last_name
FROM HR.employees E1 LEFT JOIN HR.employees E2 ON E1.manager_id = E2.employee_id;
```

Query Result x

    SQL | All Rows Fetched: 107 in 0,006 seconds

	EMPLOYEE_ID	LAST_NAME	EMPLOYEE_ID_1	LAST_NAME_1
1	168	Ozer	148	Cambrault
2	169	Bloom	148	Cambrault
3	170	Fox	148	Cambrault
4	171	Smith	148	Cambrault
5	172	Bates	148	Cambrault
6	173	Kumar	148	Cambrault
7	103	Hunold	102	De Haan

16. Сопоставить города со странами в Азии

SELECT C.country_name, L.city, R.region_name FROM HR.countries C LEFT OUTER JOIN HR.locations L ON L.country_id = C.country_id LEFT OUTER JOIN HR.regions R USING(region_id)WHERE R.region_name = 'Asia';

Worksheet		Query Builder
		<pre>SELECT C.country_name, L.city, R.region_name FROM HR.countries C LEFT OUTER JOIN HR.locations L ON L.country_id = C.country_id LEFT OUTER JOIN HR.regions R USING(region_id)WHERE R.region_name = 'Asia';</pre>
Query Result		x
		All Rows Fetched: 7 in 0,004 seconds
COUNTRY_NAME	CITY	REGION_NAME
1 Australia	Sydney	Asia
2 China	Beijing	Asia
3 India	Bombay	Asia
4 Japan	Tokyo	Asia
5 Japan	Hiroshima	Asia
6 Singapore	Singapore	Asia
7 Malaysia	(null)	Asia

17. Используйте синоним синтаксиса LEFT JOIN, чтобы получить точно такие же результаты, как в примере выше.

SELECT C.country_name, L.city, R.region_name FROM HR.countries C LEFT JOIN HR.locations L ON L.country_id = C.country_id LEFT JOIN HR.regions R USING(region_id)WHERE R.region_name = 'Asia';

Worksheet		Query Builder
		<pre>SELECT C.country_name, L.city, R.region_name FROM HR.countries C LEFT JOIN HR.locations L ON L.country_id = C.country_id LEFT JOIN HR.regions R USING(region_id)WHERE R.region_name = 'Asia';</pre>
Query Result		x
		All Rows Fetched: 7 in 0,004 seconds
COUNTRY_NAME	CITY	REGION_NAME
1 Australia	Sydney	Asia
2 China	Beijing	Asia
3 India	Bombay	Asia
4 Japan	Tokyo	Asia
5 Japan	Hiroshima	Asia
6 Singapore	Singapore	Asia
7 Malaysia	(null)	Asia

18. Укажите в каждом отделе номер сотрудника и фамилию руководителя, включая отделы без руководителя.

SELECT D.department_name, D.manager_id, E.last_name FROM HR.departments D LEFT JOIN HR.employees E ON D.manager_id = E.employee_id;

Worksheet

Query Builder

```
SELECT D.department_name, D.manager_id, E.last_name FROM HR.departments D
LEFT JOIN HR.employees E ON D.manager_id = E.employee_id;
```

Query Result x

    SQL | All Rows Fetched: 27 in 0,003 seconds

	DEPARTMENT_NAME	MANAGER_ID	LAST_NAME
1	Public Relations	204	Baer
2	Shipping	121	Fripp
3	Finance	108	Greenberg
4	Marketing	201	Hartstein
5	Accounting	205	Higgins
6	IT	103	Hunold
7	Executive	100	King

19. Протестируйте естественные объединения двух таблиц, используя данные HR.

Соедините таблицы регионов и стран с предложением WHERE, используя следующий запрос:

SELECT R.REGION_NAME, C.COUNTRY_NAME FROM REGIONS R, COUNTRIES C WHERE R.REGION_ID = C.REGION_ID ORDER BY R.REGION_NAME, C.COUNTRY_NAME;

Worksheet	Query Builder
<pre>SELECT R.REGION_NAME, C.COUNTRY_NAME FROM HR.REGIONS R, HR.COUNTRIES C WHERE R.REGION_ID = C.REGION_ID ORDER BY R.REGION_NAME, C.COUNTRY_NAME;</pre>	
Query Result x	
All Rows Fetched: 25 in 0,003 seconds	
REGION_NAME	COUNTRY_NAME
1 Americas	Argentina
2 Americas	Brazil
3 Americas	Canada
4 Americas	Mexico
5 Americas	United States of America
6 Asia	Australia

20. Протестируйте естественные объединения трех таблиц, используя данные HR.

Соедините таблицы регионов, стран и местоположений с предложением WHERE, используя следующий запрос:

SELECT R.REGION_NAME, C.COUNTRY_NAME, L.CITY FROM HR.REGIONS R, HR.COUNTRIES C, HR.LOCATIONS L WHERE R.REGION_ID = C.REGION_ID AND C.COUNTRY_ID = L.COUNTRY_ID ORDER BY R.REGION_NAME, C.COUNTRY_NAME, L.CITY;

Worksheet Query Builder

```
SELECT R.REGION_NAME, C.COUNTRY_NAME, L.CITY
FROM HR.REGIONS R, HR.COUNTRIES C, HR.LOCATIONS L
WHERE R.REGION_ID = C.REGION_ID AND C.COUNTRY_ID = L.COUNTRY_ID
ORDER BY R.REGION_NAME, C.COUNTRY_NAME, L.CITY;
```

Query Result x

    SQL | All Rows Fetched: 23 in 0,005 seconds

	REGION_NAME	COUNTRY_NAME	CITY
1	Americas	Brazil	Sao Paulo
2	Americas	Canada	Toronto
3	Americas	Canada	Whitehorse
4	Americas	Mexico	Mexico City
5	Americas	United States of America	Seattle
6	Americas	United States of America	South Brunswick

21. Протестируйте NONEQUIJOINS используя данные HR

Выберите сотрудников, чья зарплата меньше минимальной зарплаты на других должностях, и перечислите должности, используя следующий запрос:

SELECT E.LAST_NAME, E.SALARY, J.MIN_SALARY, J.JOB_TITLE FROM HR.EMPLOYEES E JOIN HR.JOBS J ON (E.SALARY < J.MIN_SALARY) ORDER BY J.JOB_TITLE, E.SALARY, E.LAST_NAME;

Worksheet Query Builder

```
SELECT E.LAST_NAME, E.SALARY, J.MIN_SALARY, J.JOB_TITLE FROM HR.EMPLOYEES E
JOIN HR.JOBS J ON (E.SALARY < J.MIN_SALARY)
ORDER BY J.JOB_TITLE, E.SALARY, E.LAST_NAME;
```

Query Result x

All Rows Fetched: 991 in 0,023 seconds

	LAST_NAME	SALARY	MIN_SALARY	JOB_TITLE
1	Olson	2100	4200	Accountant
2	Markle	2200	4200	Accountant
3	Philtanker	2200	4200	Accountant
4	Gee	2400	4200	Accountant
5	Landry	2400	4200	Accountant
6	Colmenares	2500	4200	Accountant

22. Выберите все отделы и фамилии сотрудников в этих отделах. Включите отделы без сотрудников:

SELECT D.DEPARTMENT_NAME, E.LAST_NAME FROM HR.DEPARTMENTS D LEFT OUTER JOIN HR.EMPLOYEES E ON (D.DEPARTMENT_ID = E.DEPARTMENT_ID) ORDER BY D.DEPARTMENT_NAME, E.LAST_NAME;

Worksheet Query Builder

```
SELECT D.DEPARTMENT_NAME, E.LAST_NAME FROM HR.DEPARTMENTS D
LEFT OUTER JOIN HR.EMPLOYEES E ON (D.DEPARTMENT_ID = E.DEPARTMENT_ID)
ORDER BY D.DEPARTMENT_NAME, E.LAST_NAME;
```

Query Result x

All Rows Fetched: 122 in 0,006 seconds

	DEPARTMENT_NAME	LAST_NAME
1	Accounting	Gietz
2	Accounting	Higgins
3	Administration	Whalen
4	Benefits	(null)
5	Construction	(null)
6	Contracting	(null)

23. Выберите фамилии всех сотрудников и их отделов. Включите сотрудников без отдела:

SELECT D.DEPARTMENT_NAME, E.LAST_NAME FROM HR.DEPARTMENTS D RIGHT OUTER JOIN HR.EMPLOYEES E ON (D.DEPARTMENT_ID = E.DEPARTMENT_ID) ORDER BY D.DEPARTMENT_NAME NULLS FIRST, E.LAST_NAME;

Worksheet Query Builder

```
SELECT D.DEPARTMENT_NAME, E.LAST_NAME FROM HR.DEPARTMENTS D
RIGHT OUTER JOIN HR.EMPLOYEES E ON (D.DEPARTMENT_ID = E.DEPARTMENT_ID)
ORDER BY D.DEPARTMENT_NAME NULLS FIRST, E.LAST_NAME;
```

Query Result x

All Rows Fetched: 107 in 0,005 seconds

	DEPARTMENT_NAME	LAST_NAME
1	(null)	Grant
2	Accounting	Gietz
3	Accounting	Higgins
4	Administration	Whalen
5	Executive	De Haan
6	Executive	King

24. Выберите все отделы и фамилии всех сотрудников. Включите отделы без сотрудников и сотрудников без отделов:

SELECT D.DEPARTMENT_NAME, E.LAST_NAME FROM HR.DEPARTMENTS D FULL OUTER JOIN HR.EMPLOYEES E ON (D.DEPARTMENT_ID = E.DEPARTMENT_ID) ORDER BY D.DEPARTMENT_NAME NULLS FIRST, E.LAST_NAME;

local.sql x local x DEPARTMENTS x

SQL Worksheet History

Worksheet Query Builder

```
SELECT D.DEPARTMENT_NAME, E.LAST_NAME FROM HR.DEPARTMENTS D
FULL OUTER JOIN HR.EMPLOYEES E ON (D.DEPARTMENT_ID = E.DEPARTMENT_ID)
ORDER BY D.DEPARTMENT_NAME NULLS FIRST, E.LAST_NAME;
```

Query Result x

SQL | All Rows Fetched: 123 in 0,006 seconds

	DEPARTMENT_NAME	LAST_NAME
1	(null)	Grant
2	Accounting	Gietz
3	Accounting	Higgins
4	Administration	Whalen
5	Benefits	(null)
6	Construction	(null)

25. Выберите фамилию сотрудников без отдела:

SELECT E.LAST_NAME FROM HR.EMPLOYEES E LEFT OUTER JOIN HR.DEPARTMENTS D ON
(E.DEPARTMENT_ID = D.DEPARTMENT_ID) WHERE D.DEPARTMENT_ID IS NULL;

Worksheet Query Builder

```
SELECT E.LAST_NAME FROM HR.EMPLOYEES E LEFT OUTER JOIN HR.DEPARTMENTS D
ON (E.DEPARTMENT_ID = D.DEPARTMENT_ID) WHERE D.DEPARTMENT_ID IS NULL;
```

Query Result x

SQL | All Rows Fetched: 1 in 0,002 seconds

	LAST_NAME
1	Grant